

Restricting resolutions: effect-driven semantics of numerical approximation

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Abstract. Approximators such as *approximately* or *exactly* have been analyzed either as operators that manipulate a numeral’s denotation, or as devices that select a granularity level at which the numeral is interpreted. A single numeral, however, can host several approximating devices at once. In Ukrainian, an overt adverb, a preposition, an indefinite, and approximative inversion may co-occur with one numeral, yet the result conveys the same interval as any one alone, differing only in an intensified hedging effect. Pragmatic halo theories correctly predict that imprecision in such cases does not compound, but fail to rule out mixed-precision stacks like *approximately exactly*; granularity-based accounts predict that stacking is infelicitous altogether. I propose an approach where a single mechanism derives the idempotence of same-precision stacking alongside the phenomena prior views miss. I argue that a bare numeral introduces a family of resolution descriptions—the precision levels at which it may be read—and approximators function like indefinites and definites over this family: *approximately* restricts it to its non-exact members without choosing among them, and *exactly* retrieves the unique exact member. Stacking is then intersection of conditions, idempotent for same-precision devices and contradictory for mixed-precision ones. I implement the account in effect-driven semantics and lift it to probabilistic dynamic semantics. Treating the choice of resolution as uncertain yields an interpretation peaked at the numeral; roundness and magnitude phenomena then reduce to how the prior over resolutions varies with it. The mixed-precision clash is rescued in discourse as an instance of ordinary anaphora.

Keywords. approximation; imprecision; numerals; granularity; effect-driven semantics; probabilistic dynamic semantics.

1. Introduction. Numerical expressions may be used with varying levels of precision. Depending on the context, terms that have a precise meaning can be used imprecisely:

- (1) A cup of coffee costs five dollars.

Suppose a traveler is complaining that coffee in the city is expensive. If the actual price is \$4.90, the utterance of (1) is still felicitous and counts as true, interpreted loosely. Now suppose a barista utters (1) to answer a customer asking for the price. Here the numeral is interpreted precisely: if the coffee in fact costs \$4.90, the same sentence is now judged false.

Aside from contextual clues, the level of precision can be regulated by means of various *approximators*: devices that signal the numerical expression should be interpreted more or less precisely:

- (2) a. A cup of coffee costs exactly five dollars.
b. A cup of coffee costs around five dollars.

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Uttering (2a) forces the most precise reading, where *exactly* is understood to refer to the exact price mentioned. Conversely, using a modifier such as *around* in (2b) invokes an interval-like reading for the numeral, such that the sentence would still be interpreted as truthful even if the price does not exactly equal the numeral uttered.

To understand the underlying semantics of these modifiers—whether they manipulate the boundaries of the numeral’s denotation or set the level of precision at which the numeral should be interpreted—one diagnostic is to observe their combinatorial behavior. What happens when multiple approximators modify the same numeral? When multiple imprecise devices are stacked, does the imprecision compound, representing a progressively wider interval, or is the second device simply redundant? Furthermore, when a precise and an imprecise device are mixed, such as in *exactly approximately*, do these operations sequentially overwrite one another, cancel out, or trigger a semantic clash?

Observing this combinatorics in English, however, is difficult, as approximative devices do not seem to combine freely. In English, stacking two approximators onto a single numeral is generally degraded out of the blue: *approximately around five dollars* and *exactly precisely five dollars* are both odd. So are mixed combinations such as *exactly approximately five dollars* or *approximately exactly five dollars*. Based on this restricted distribution, one might conclude that approximators categorically resist co-occurrence on semantic grounds (cf. Sauerland & Stateva 2011, who make such prediction for a subset of approximators they call *scalar*, which includes *approximately* and *exactly*). But this creates a confound: English obscures whether this resistance reflects a semantic incompatibility, or a morphosyntactic constraint against modifying a numeral with multiple similar adverbs.

Fortunately, the inventory of approximating devices extends well beyond simple modifiers such as the ones mentioned, and their realization varies significantly across languages.¹ This variability serves as a theoretical probe. In a language where such devices belong to diverse syntactic categories, they can co-occur, allowing us to observe the semantic combinatorics of approximation directly. To treat this as a general diagnostic, however, we must assume that structurally distinct devices share the same semantic core. My working hypothesis is thus that languages differ in their inventories of approximating devices, not in the underlying semantics of approximation.

Ukrainian presents exactly this kind of landscape. Alongside overt modifiers such as *pry-blyzno* ‘approximately’ (3a), it uses other strategies to express imprecision, such as approximative prepositional constructions (3b), or approximative inversion, in which the canonical order of the numeral and the noun is inverted, yielding approximative meaning (3c).

- (3) a. Roman kupyv prybyzno sim knyh.
 Roman bought approximately seven books-GEN.PL
 ‘Roman bought approximately seven books.’
- b. Roman kupyv zo sim knyh.
 Roman bought APPRX seven books-GEN.PL
 ‘Roman bought approximately seven books.’

¹ English, for example, employs hedging constructions such as *or so*, *more or less*, *-ish*, which exhibit approximator-like behavior and pattern differently syntactically from approximative modifiers. These are permitted to stack in certain combinations. See Ferson et al. (2015) for a more comprehensive list of English approximators, and Channell (1994) for discussion of approximator stacking in English.

- c. Roman kupyv knyh sim.
 Roman bought books-GEN.PL seven
 ‘Roman bought approximately seven books.’

Perhaps unexpectedly, these are allowed to co-occur:

- (4) a. Roman kupyv pryblyzno knyh sim.
 Roman bought approximately books-GEN.PL seven
 ‘Roman bought approximately seven books.’
- b. Roman kupyv knyh zo sim.
 Roman bought books-GEN.PL APPRX seven
 ‘Roman bought approximately seven books.’
- c. Roman kupyv pryblyzno knyh zo sim.
 Roman bought approximately books-GEN.PL APPRX seven
 ‘Roman bought approximately seven books.’

The stacked forms in (4), crucially, do not convey a wider or narrower interval than any single device would. Uttering any of them expresses the same sense of approximation. The imprecision does not compound, and the utterances differ only in a heightened sense of the speaker’s hedging. Stacking as such looks semantically idempotent, but pragmatically additive.

Consider what the main branches of existing analyses of imprecision predict for this pattern. Under a pragmatic theory of imprecision (Lasnik 1999), where a denotation assigned to an expression of a language is associated with a *pragmatic halo*, approximators either tighten the halo of an expression or expand its denotation to include all values in said halo. Conversely, a family of approaches based on the notion of a *granularity scale* (Krifka 2009; Sauerland & Stateva 2011; Solt 2015; Gyarmathy 2017) models approximators as granularity setters, or functions that map points to intervals around them bounded by the granularity level. It turns out that both analyses make similar predictions in regard to stacking of approximative devices. If approximators operate on pragmatic halos of numerical expressions by actively expanding a precise denotation, using multiple low-precision approximators is predicted to be strictly vacuous. Similarly, on the granularity scale approach, once the inner approximator sets the resolution parameter, subsequent scalar approximators in the scope of the first are rendered vacuous and therefore infelicitous.

Beyond same-class stacking, approximators exhibit complex behavior when mixing precision levels. For instance, stacking a precise and an imprecise device (*exactly approximately 5 o’clock*) is odd when uttered out of the blue. However, the acceptability of such mixed stacking improves if the inner phrase (*approximately 5 o’clock*) has already been established as a salient discourse referent (Zaroukian 2013). Solt (2015) notes a similar effect involving comparatives, observing that constructions such as *more than about 100* are licensed if *about 100* exists as a referent in the prior context.

The Ukrainian data show that same-precision stacking is felicitous, contrary to the granularity-based prediction, and conveys the same interval as a single device. Pragmatic halo theories predict this non-compounding correctly, but they overgenerate in the mixed-precision cases: if both devices simply manipulate the halo, nothing rules out *approximately exactly*, yet such a stack is

infelicitous out of the blue and acceptable only once the inner phrase is discourse-given. I propose an account that derives both patterns from one mechanism. The same treatment of approximators that makes same-precision stacking non-compounding yields the mixed-precision clash and its discourse rescue.

I argue that approximators function neither as denotation-wideners nor as granularity-setters, but as expressions that pick out resolutions in the way that indefinites and definites pick out individuals. A bare numeral, on this account, introduces a family of candidate resolutions—the levels of precision at which it may be read—and leaves the choice among them open. An imprecise approximator restricts this family to its non-exact members, without choosing among them, mirroring an indefinite; a precision device retrieves the unique exact member, as a definite does. I present the Ukrainian stacking data in more detail in Section 2 and situate this account against prior theories of imprecision in Section 3, before formalizing the architecture in Section 4.

Because both kinds of devices contribute conditions on the same family, stacking them amounts to intersection, and stacking identical conditions is idempotent. This predicts the behavior observed in (4): the width of the interval remains untouched under stacking. The heightened sense of hedging contributed by the stacked form does not arise from its truth-conditional content, which is the same as with a single device. Instead, it comes from the form of the utterance: the speaker uses several devices where one would have sufficed, and the listener infers that this longer form has a reason and may conclude that the speaker is signaling extra uncertainty. Mixed-precision stacks, on the other hand, impose mutually exclusive conditions on a shared target, resulting in a clash.

I implement the analysis in the effect-driven framework of Bumford & Charlow (2026), in which indefinites and definites are treated as carrying computational effects, and lift it to probabilistic dynamic semantics (PDS; Grove & White 2024). The dynamic component offers a principled solution to the mixed-precision puzzle: an approximated phrase, once asserted, introduces a discourse referent, and a later occurrence can refer back to it rather than introduce a fresh one. Section 5 models this phenomenon as an instance of ordinary anaphora. The probabilistic component lets us treat the choice of resolution as uncertain, without altering the underlying compositional semantics. Marginalizing over that uncertainty yields an interpretation peaked at the numeral, as I show in Section 6. Section 7 concludes.

2. Data. In this section, I present the Ukrainian data in more detail, contrasting it with English where possible, as well as outline some additional puzzles that concern approximator usage. As noted above, I suggest that languages differ in their inventory of approximating devices, not in the semantics of approximation.

2.1. APPROXIMATIVE DEVICES IN UKRAINIAN. Ukrainian is a fusional language with rich morphology that has a wide inventory of approximating devices, ranging from overt modifiers to approximation expressed by syntactic configuration.

2.1.1. OVERT MODIFIERS. The most productive strategy for expressing an imprecise numeral meaning is to use an overt modifier with the numeral, similar to English. Examples of such approximators are *pryblizno* ‘approximately’, *orijentovno* ‘approximately’, *majže* ‘nearly’, *bilš-menš* ‘more or less’. To express a more precise meaning, modifiers such as *točno* ‘precisely’ or *rivno* ‘exactly’ are used:

- (5) a. Roman kupyv pryblyzno sim knyh.
 Roman bought approximately seven books-GEN.PL
 ‘Roman bought approximately seven books.’
- b. Roman kupyv rivno sim knyh.
 Roman bought exactly seven books-GEN.PL
 ‘Roman bought exactly seven books.’

In addition, Ukrainian permits indefinites such as *des* ‘somewhere’ or *ščos* ‘something’ to function as markers of approximation:

- (6) a. Roman kupyv des’ sim knyh.
 Roman bought somewhere seven books-GEN.PL
 ‘Roman bought approximately seven books.’
- b. Roman kupyv ščos’ sim knyh.
 Roman bought something seven books-GEN.PL
 ‘Roman bought approximately seven books.’

The spatial/referential indefiniteness in such contexts shifts to mark quantity or degree imprecision. Approximative usage of indefinites is not unique to Ukrainian, as English exhibits similar behavior with *somewhere around* or *something like*. However, there are potentially important syntactic differences between languages: Ukrainian allows using the bare single-word indefinite, as shown in (6), whereas English syntactically requires additional hedges such as *around* or *like* to felicitously use the indefinite:

- (7) a. Roman bought somewhere around seven books.
 b. *Roman bought somewhere seven books.
 c. Roman bought something like seven books.
 d. *Roman bought something seven books.

Because English *somewhere* cannot be used as an independent approximator, I do not, for the purposes of this paper, consider English structures such as *somewhere around* instances of approximator stacking. On the contrary, since Ukrainian indefinites *des* and *ščos* do function as approximators independently, I assume they can be considered separate elements in Ukrainian stacked constructions of the form *des’ pryblyzno*.

2.1.2. PREPOSITIONAL APPROXIMATION. Certain prepositions can also give rise to approximative readings, such as *kolo* ‘around’, *bilja* ‘around’, and *blyz’ko* ‘near’:

- (8) a. Roman kupyv kolo semy knyh.
 Roman bought around seven-GEN books-GEN.PL
 ‘Roman bought approximately seven books.’

All of them require genitive case marking on the object and have a quantificational meaning only if the object is itself a quantifier. Otherwise, the interpretation is merely proximative.

In addition, approximation in Ukrainian can also be expressed by using the preposition *z* (or its euphonic variants *zo*, *iz*) with the accusative case marking on the object:

- (9) a. Roman kupyv zo sim knyh.
 Roman bought APPRX seven-ACC books-GEN.PL
 ‘Roman bought approximately seven books.’

This usage is distinct from the regular genitive-governing *z* ‘from’, typically used to denote location, as well as instrumental-governing *z* ‘with’. Historically, prepositions such as *na* ‘on’ and *v* ‘in’ have been noted to be used in a similar sense with the accusative (Shevelov 1963). However, in modern Ukrainian, those are almost never used to approximate.

I am not aware of existing studies on the specifics of how *z*+ACC functions in modern standard Ukrainian; and while the phenomenon is attested in other Slavic languages (Billings 1995), note that Ukrainian seems to exhibit different properties, such as not generally adhering to the single-word complement restriction, warranting further exploration. For the purposes of this work, I view *z*+ACC as another approximator construction, staying agnostic to the syntactic intricacies of its use.

2.1.3. APPROXIMATIVE INVERSION. *Approximative inversion* is a construction in Ukrainian (and other East Slavic languages) where the canonical numeral-noun order is reversed, yielding an approximative reading:

- (10) a. Roman kupyv sim knyh.
 Roman bought seven books-GEN.PL
 ‘Roman bought seven books.’
 b. Roman kupyv knyh sim.
 Roman bought books-GEN.PL seven
 ‘Roman bought about seven books.’

Most of the existing research on approximative inversion has been focused on its syntax, with semantics receiving far less attention. Zaroukian (2012) claims it is a marker of epistemic uncertainty, used in contexts where the speaker is necessarily uncertain about the exact value of the numeral. Under such view, approximative inversion licenses a null modal operator *maybe*, which takes scope over the numeral. Khrizman & Rothstein (2015) refute this proposal, providing examples where the speaker has full world knowledge, but still can felicitously approximate by using an inverted construction. In turn, they suggest that approximative inversion functions as a measure construction involving estimation. Approximation effects are then derived by using an estimation operation which maps entities onto a range of numerals centered around a specific number, rather than onto a single precise value. Finally, Sauerland & Stateva (2011) briefly mention approximative inversion as an instance of an ordinary scalar approximator. On their granularity scale view, the inverted construction licenses a silent counterpart of approximative *some*, suggesting it has identical semantic contribution.

Similarly to the *z*+ACC construction, I am not aware of any studies on the specifics of approximative inversion in modern standard Ukrainian. Notably, approximative inversion in Ukrainian appears to have a different profile compared to previous accounts, as it extends to domains they don't predict, such as ordinals and indefinite quantifiers. In this paper, I again remain agnostic about the specifics of the meaning of approximative inversion, viewing it as an approximator on the same terms as other approximative devices mentioned.

2.2. APPROXIMATOR COMBINATION. Despite there being considerable research on the specifics of imprecision and approximation, the idea that they can felicitously stack with each other has received little attention. Channell (1994), for example, does not outright deny the possibility of approximator stacking in English, but suggests that it is infrequent and restricted to certain combinations of overt modifiers and approximative expressions involving the use of *or*. At the same time, the general unavailability of stacking mixed-precision approximators such as *exactly approximately* or *approximately exactly* in English, as well as blocked same-precision combinations such as *about approximately* (Sadock 1977) has led, e.g., Sauerland & Stateva (2011) to predict scalar approximators cannot be stacked whatsoever. I argue that this prediction is not entirely correct in view of novel Ukrainian data.

2.2.1. FELICITOUS STACKING. First, recall (4), which shows Ukrainian allows stacking multiple approximators hosted by the same numeral. Extreme cases, such as stacking one device of each of the above classes, are similarly felicitous:

- (11) Roman kupyv des' pryblyzno knyh zo sim.
 Roman bought somewhere approximately books-GEN.PL APPRX seven
 'Roman bought approximately seven books.'

Crucially, utterances such as (11), where approximation is expressed in four different ways at the same time, result in the same interval interpretation as if only one of them had been used. One way to verify this is the following diagnostic: if the stacked form conveyed a different (e.g., wider) interval compared to the single form, it should be possible to use it as a correction. However, rejecting the single form to assert the stacked form is judged infelicitous, as speaker B's response in (12) expresses no actual disagreement:

- (12) a. A: Roman kupyv pryblyzno sim knyh.
 Roman bought approximately seven books-GEN.PL
 'Roman bought approximately seven books.'
- b. B: #Ni, vin kupyv des' pryblyzno knyh zo sim.
 no he bought somewhere approximately books-GEN.PL APPRX seven
 'No, he bought approximately seven books.'

By contrast, a genuine correction (e.g., *Ni, vin kupyv pryblyzno visim knyh* 'No, he bought approximately eight books') is felicitous. Note that B's response in (12b) is not degraded merely because it repeats the numeral. Correcting with a precision device, as in *Ni, vin kupyv rivno sim knyh* 'No, he bought exactly seven books', is likewise felicitous, since it genuinely alters the truth conditions. Because any scenario verifying the single device also verifies the stacked configuration and vice versa, the forms are mutually entailing. The imprecision does not compound. This

is surprising under a compositional view where every approximator contributes a separate meaning of imprecision—if that were the case, then the resulting interpretation would denote a wider interval than each of them separately.

Using two or three approximators instead of just one is common in modern standard Ukrainian, and stacked forms are readily attested in corpora. The following examples from GRAC v.19a (Shvedova et al. 2017–) each combine three devices: the indefinite *des*’ with the adverb *prybyl’no* and an inverted measure noun in (13a), and *des*’ with inversion and the preposition *zo* in (13b):

- (13) a. Ce *des*’ *prybyl’no* *kilometriv* *desjat*’.
 this somewhere approximately kilometers-GEN.PL ten
 ‘It is somewhere around ten kilometers.’
 (GRAC: I. Prokopchuk, *Versiji*, 2022)
- b. Tak *tryvalo* *des*’ *xvylyn* *zo* *p’jat*’, *može*, *bil’she*.
 so lasted-PST.N somewhere minutes-GEN.PL APPRX five maybe more
 ‘It went on for approximately five minutes, maybe more.’
 (GRAC: Ju. Izdryk, *Ostriv KRK*, 1998)

The stacked construction gives rise to a pragmatic effect of heightened hedging: the hearer infers that the speaker is more uncertain about the exact value, but not more imprecise in the sense of conveying a wider interval. Because a single approximator can convey either genuine ignorance or rounding for communicative efficiency, this contrast is best diagnosed through an evidential context test. In a scenario where the speaker possesses perfect evidence and is merely summarizing, the single form is felicitous, while the stacked form is degraded. Suppose the speaker has just checked the receipt, which lists exactly 68 books, and is summarizing the purchase for a friend:

- (14) a. *Roman kupyv prybyl’no simdesjat knih*.
 Roman bought approximately seventy books-GEN.PL
 ‘Roman bought approximately seventy books.’
- b. #*Roman kupyv des*’ *prybyl’no* *knih* *zo* *simdesjat*.
 Roman bought somewhere approximately books-GEN.PL APPRX seventy
 ‘Roman bought approximately seventy books.’

Conversely, in a context of poor evidence, where the speaker genuinely does not know the exact number, both forms are felicitous. If the speaker uses several devices where one would have sufficed, the hearer may infer that this longer form has a reason: signaling uncertainty about the exact value. The interval itself is untouched.

The pragmatic effect, however, is most pronounced in extreme cases such as (11) or (13b), whereas (4a) or (4b), for example, are less likely to result in a more uncertain interpretation compared to (3a) and (3b) respectively. I attribute this to frequency: stacks of two devices are common enough to have lost their marked character, while larger stacks remain noticeably long-winded.

2.2.2. THE SAME-CLASS RESTRICTION. I have established that approximators can indeed stack felicitously. However, this is too broad of a generalization: it is also apparent that not all approximators stack. For example, similar to English *approximately around*, Ukrainian forbids stacking two or more approximating devices like the ones in the following examples:

- (15) a. *Roman kupyv pryblyzno orijentovno sim knyh.
 Roman bought approximately approximately seven books-GEN.PL
 intended: ‘Roman bought approximately seven books.’
- b. *Roman kupyv des’ ščos’ sim knyh.
 Roman bought somewhere something seven books-GEN.PL
 intended: ‘Roman bought approximately seven books.’
- c. *Roman kupyv kolo bilja semy knyh.
 Roman bought around around seven-GEN books-GEN.PL
 intended: ‘Roman bought approximately seven books.’

Note that one common thing in the examples from (15) is that stacking is judged unacceptable when the devices that stack share some morphosyntactic features; that is, combinations of two overt modifiers, two approximative prepositions, or two indefinites are blocked, while mixing them seems to work without issue. I follow Channell (1994), who derives a parallel restriction for approximator combinations in English, in assuming this to be a morphosyntactic constraint on the realization of approximative elements, not a semantic one. Because my primary concern is the underlying meaning, I abstract away from the formal mechanics of this syntactic restriction.

2.2.3. MIXED-PRECISION CLASH AND RESCUE. Mixed-precision stacking presents an additional puzzle: combining a precise and an imprecise device is generally judged infelicitous when uttered out of the blue:

- (16) a. #This road spans approximately exactly 50 kilometers.
 b. #This road spans exactly approximately 50 kilometers.

However, as Zaroukian (2013) points out, these combinations can be felicitous in the right context. Under this account, expressions like *exactly 50 kilometers* or *approximately 50 kilometers* may denote a prototype for the notion of being exactly or approximately 50 kilometers long, respectively. Uttering (16a) then asserts the road is reasonably close to this ideal exact-length representation, whereas (16b) describes a road that is very close to the ideal approximate prototype. I take the core intuition to be that the inner expression is not interpreted afresh: it picks up a referent already established in the discourse, one that carries its own, already-fixed level of precision, and the outer approximative device operates on that referent.

The idea that approximated expressions somehow interact with the discourse state is further supported by a similar observation Solt (2015) makes about comparatives. Overt approximators are considered ill-formed in the positive comparative sentences such as the following:

(17) #This road spans more than about 50 kilometers.

However, as Solt points out, expressions such as (17) are acceptable in reply contexts; if *about 50 kilometers* was already established as a salient discourse referent by some previous move, one can utter (17), for example, to refute the prior claim about the length of the road.

The only formal account of this clash-and-rescue phenomenon I am aware of is presented by Zaroukian (2013), who proposes a static coerced-reading analysis that relies on introducing a specific degree type-shifting rule for inner approximator constructions. I argue that a dynamic semantics approach fares better at explaining such cases and make this precise in Section 5.

2.3. OTHER PUZZLES. Multiple other observations have been made about numerical expressions used approximately. Out of those, I note a couple of the most prominent facts that a theory of approximators should provide an explanation for.

2.3.1. THE ROUNDNESS AND MAGNITUDE EFFECTS. First, there is a contrast in interpretation between round and non-round numbers (Krifka 2009). Numerical expressions we interpret as round, such as *one hundred*, *thirty*, *six o'clock*, *forty-five minutes* favor an approximate interpretation, while non-round numbers such as *one hundred and two*, *thirty-eight*, *six thirteen*, or *forty-two minutes* invoke a sense of higher precision. Interestingly for the purposes of this paper, the roundness effect persists when the relevant numbers are used with approximators:

(18) How many more kilometers do we have to drive?

- a. Around one hundred.
- b. Around ninety-eight.

Uttering (18a) seems to invoke a wider sense of interval than (18b); that is, one can arguably felicitously answer *around one hundred* when the actual distance is in the [90, 110] interval, whereas *around ninety-eight* is more likely to have a more precise interpretation (though still approximate), such as [95, 100].

Second, Lefort et al. (2016, 2017) point out that the relative magnitude of the approximated numeral also plays a part in identifying the boundaries of the interval that is being denoted. As an example, the authors suggest that the interval of *about 100* is intuitively biased towards a narrower interpretation than the interval of *about 800*. Similar predictions are made by Sadock (1977), who suggests that the magnitude of the number alters how wide the interval is perceived; and by Channell (1994), who argues in favor of the roundness contrast. Both observations are also supported by the experimental findings of Ferson et al. (2015), who show that both roundness and order of magnitude are good predictors of the related intervals. Hence, I suggest that the final reading of the approximated expression depends on contributions from both the specific numeral and the approximator.

2.3.2. THE *around-between* CONTRAST. In a recent line of work, Hénot-Mortier et al. (2026) make a different observation about approximators specifically, as compared to similar expressions that force an interval reading. They look at the difference in constructions using *around* and *between*:

- (19) a. There were around 70 people at the party.
 b. There were between 60 and 80 people at the party.

Intuitively, both can be used by the speaker to describe a situation in which they are uncertain about the specific number of attendees. Hénót-Mortier et al. point out they are not totally synonymous, however. First, they suggest that *around* does not specify a sharp interval, whereas *between* does. Second, the authors note that *around* conveys some extra information about what the most probable numbers are in this case; specifically, those are the values closer to the uttered numeral. *Between*-expressions, on the other hand, result in a uniform reading of the conveyed interval. The values outside of the interval are excluded, but all numbers within the interval are equally probable, conveying sheer uncertainty. Hénót-Mortier et al. model this peakedness probabilistically; I adopt a probabilistic treatment for the same reason.

3. Theories of imprecision. Two accounts of imprecision emerged as the main branches for explaining how approximators function. Under a view rooted in pragmatics, language is taken to be precise by default, which leads to us saying things that are literally false, but interpreted as true based on the context. The alternate idea is that language allows imprecise statements that are true nonetheless. Here, I present both accounts briefly, focusing specifically on their treatment of approximative devices; for a more in-depth explanation, I refer the interested reader to Zaroukian (2013) and Solt (2015).

3.1. **PRAGMATIC HALOS.** Speakers very often utter sentences which are not, strictly speaking, true. Lasersohn (1999) addresses this phenomenon as instances of pragmatic slack; while not originally designed as a theory of vagueness, it is often treated as such (see, for example, Burnett 2012). Expressions are treated to have only a precise meaning, and a denotation is then associated with a **pragmatic halo**—a set of entities that differ from the denotation in ways that are pragmatically ignorable. Even if the exact expression is then false, the felicitousness of its utterance is based on whether some element in its halo is true. Using an imprecise, approximated numeral construction as such leads to uttering a false expression that is nonetheless close enough to the truth, so it is interpreted as being true.

Lasersohn analyzes *loosely speaking* and *roughly*—I expect approximators such as *approximately* or *about* to receive the same treatment—as having a function of expanding the denotation of an expression to include all values in its halo. Words that signal a high level of precision, such as *exactly*, tighten the halo of an expression. Note the functions of two kinds of approximators are not entirely parallel. *Exactly* is considered truth-conditionally vacuous, as its only function is to tighten the amount of allowed slack. In contrast, *about* expands the literal denotation to encompass the values already present within the halo of the modified expression, without widening the halo itself (see Lauer 2012 for a more in-depth analysis). Because all expressions are considered to be semantically precise, imprecision is treated as a purely pragmatic phenomenon.

Under the pragmatic halos approach, halos of complex expressions are derived compositionally based on the elements in the halos of their constituents. The combination of different approximators, if able to avoid a type mismatch, then should result in a vacuous update, which I discuss in more detail below.

3.2. **GRANULARITY SCALES.** An alternate family of approaches to modeling imprecision is based on the notion of a **granularity scale** (Krifka 2009; Sauerland & Stateva 2011; Solt 2015;

Gyarmathy 2017). The shared idea is that a numerical expression is interpreted relative to a contextually active level of granularity, where scales differ in coarseness according to the size of their minimal units. For example, length might be reported on the basis of scales counting in units, tens, hundreds of meters; time might be measured in minutes, five-minute intervals, quarter hours, half hours, hours, and so forth. Numerals then live as salient points on those scales, and the interpretation of a specific numerical expression depends on whichever scale is active. On a fine scale, the numeral receives a precise interpretation; on a coarse one, each unit covers a wider range, licensing approximation. On this view, imprecision is not a matter of asserting a falsehood that is close enough to the truth, but of evaluating an expression at a coarser resolution at which it is literally true.

Scale-based accounts differ in how granularity is formally represented. Krifka (2009) uses a family of entirely distinct scales, aligned on the basis of their granularity levels. Sauerland & Stateva (2011) treat granularity as a contextual parameter of interpretation, formalized as a granularity function that maps points of the scale to an interval containing it, partitioning the scale into cells whose members are indistinguishable at that resolution. Solt (2015) recasts granularity in terms of sets of alternatives to measure expressions, with the relevant alternatives fixed by a contextually determined granularity level. Gyarmathy (2017) proposes that a parameter of the interpretation should be not a lone granularity function, but a set of granularity functions; under this view, a granularity level is defined as a set of granularity functions over the same scale which have the same minimally distinguishable interval size. The two points on a scale are then considered different at a particular granularity level only if they are distinguished by all the granularity functions of that level.

Approximators, on these accounts, are devices that manipulate or exploit the active granularity. Sauerland & Stateva view words such as *approximately* as operators that set the granularity parameter to the coarsest available in context, resulting in an imprecise interpretation for the numeral; Solt has approximators associate measure expressions with the intervals whose widths are constrained by the contextually determined granularity level. On either implementation the effects of approximators run through a single selected granularity; Sauerland & Stateva take this to predict that scalar approximators cannot stack, as a second approximator restricts an already restricted parameter and is vacuous.

Accounts in this family share a principled derivation of the roundness phenomenon by rooting it in the mathematical properties of the scales themselves. A numeral can only be evaluated on a given scale if it is a salient point on that scale. A round numeral like *one hundred* is divisible by many coarse units (tens, fifties, hundreds), allowing it to be interpreted at coarse granularities, where each point spans a wider interval. Conversely, a non-round numeral like *ninety-eight* is absent from these coarse scales, forcing interpretation at a finer granularity and necessarily resulting in a narrower, more precise interval (Krifka 2009).

3.3. THEORY COMPARISON. On the pragmatic halo account, an imprecise approximator like *about* expands a precise denotation to cover its halo. Additional approximators of the same precision find the denotation already expanded with no further halo to absorb. While such a combination is in theory vacuous, it is correctly predicted to be felicitous. The pragmatic halo approach fares worse in the face of other aforementioned puzzles, however. For mixed-precision clash-and-rescue combination the following prediction is made: since *exactly* contracts the halo and *approximately* expands the denotation to cover it fully, *approximately exactly 30* is predicted

to be unproblematic out of the blue. Given that such combinations are in fact degraded, this is counterintuitive. The roundness and magnitude phenomena are also not easily derivable. It is unclear on what basis the halos of distinct measure expressions are different; and looking for such a distinction may as well lead to the notion of granularity scale. The fact that round expressions allow more approximate interpretations can be modeled by positing larger halos for them; this, however, runs into the issue of asymmetry of “pragmatic ignorability”, a fact pointed out by Solt (2015). Finally, pragmatic halos are unstructured objects. This makes it problematic for the approach to derive the peakedness of the interval conveyed by approximated expressions. Since all the values in the pragmatic halo of the expression are considered pragmatically ignorable, *about*-expressions should receive equivalent interpretation to *between*-expressions, which is not what happens in practice (Égré et al. 2023; Hénót-Mortier et al. 2026).

Approaches based on the notion of a granularity scale also yield predictions about approximator stacking that do not match what we see in the data. On Sauerland & Stateva’s view, an approximator restricts the granularity parameter to a single coarsest level. Therefore, once the outer approximator sets the parameter to a coarse scale, a second approximator in the scope of the first is rendered entirely vacuous. The authors take this to predict that scalar approximators cannot stack at all. Notably, the account treats syntactically distinct approximative strategies as semantically uniform: Sauerland & Stateva analyze approximative inversion as equivalent to English *about*. If syntactically distinct approximative devices share the same interpretive mechanism as lexical approximators, the account predicts that they cannot co-occur with overt modifiers—a prediction empirically falsified by Ukrainian data.

Alternatively, Solt suggests that expressions modified by approximators denote scalar ranges bounded by the active granularity level. The general semantics for approximators suggests that they map points to intervals around them. Based on this, applying a second approximative device is predicted to result in a type mismatch, as the inner expression now denotes a range while the outer approximator expects a point. One might attempt to resolve this mismatch either by positing a type-shifting rule, or by uniformly treating points as small or singleton intervals from the outset. However, even if the composition is repaired, the core problem persists: because the interval-building mechanism is welded directly to the interpretation of the approximator according to this theory, applying a second approximator would iteratively expand the bounds, predicting a wider interval under stacking. To avoid this widening prediction, one would have to redefine the approximator so that a second application changes nothing—for instance, as an operation that merely rounds the numeral to the cell determined by the active granularity, so that reapplying it leaves the boundaries exactly as they are. However, this amounts to decoupling the interval-building semantics from the approximating devices. If the approximator merely triggers such rounding, the interval is in fact determined by the numeral and the granularity scale alone, which is precisely the analysis I develop below.

The granularity scale approaches fare better for the roundness and magnitude phenomena, as those can be derived directly from the distribution of numerals across scales. However, similarly to the pragmatic halo theory, they do not make correct predictions on the peakedness of approximated expressions. Scalar ranges they attribute to the approximated expressions are bounded and centered on the focused numeral, but likewise unstructured further. This incorrectly suggests that all the values in the range are equally probable.

The two theories also share a core assumption. Both associate an expression with a neighborhood of values that count as equivalent for current purposes. A pragmatic halo is a general,

unstructured set of pragmatically ignorable alternatives, while a granularity scale generates its neighborhoods from a structured system of salient points. Granularity can thus be viewed as a specialized implementation of halos for domains of measurement. I take these structured, scale-generated neighborhoods as the starting point for numerals. Importantly, framing it this way allows us to remain agnostic about domains that lack scale structure. Approximation in other domains, such as approximated quotation or categorization, may instead track similarity rather than a totally ordered scale.²

Finally, note the two traditions disagree about the locus of imprecision. For Lasersohn (1999), a loose utterance is strictly false but tolerated, while on the granularity view it is true at a coarser resolution. On this point, my account sides with the latter; I also share the assumption that a numeral’s neighborhoods are structured, generated by scales in measurement domains. The approach differs from the granularity tradition in two respects. First, I decouple the interval from the approximator, arguing that a bare numeral introduces a family of *resolution descriptions*—the candidate levels of precision at which it may be read—and approximating devices restrict this family rather than building an interval, so that stacked devices compose by intersection. Second, where prior accounts select a single active granularity, I place a probability distribution over the available resolutions and marginalize over it, which is what yields the graded, peaked interpretation of approximated numerals.

4. Proposal. I now formalize this resolution-based view. The core intuition driving the proposal is that approximators function like articles over levels of precision, an analysis I implement in the effect-driven framework of Bumford & Charlow (2026). I develop this compositional layer in four steps. Section 4.1 defines the available resolution descriptions for a given numeral. Bare numerals introduce an indeterminate choice over this family of descriptions, modeled in Section 4.2. Section 4.3 develops the article analogy further, treating *exactly* as a definite that retrieves the unique exact description, and *approximately* as an indefinite restricted to the non-exact candidates. Finally, Section 4.4 shows how these devices compose. Because they add restrictions to the same indeterminate choice, stacked approximators intersect, deriving their non-compounding interpretation, the clash, and the rescue of mixed-precision devices. Throughout this section, the combinatorics remain crisp. The discourse dynamics behind the rescues are detailed in Section 5, and the probabilistic weighting is introduced in Section 6.

4.1. RESOLUTION DESCRIPTIONS. Recall example (1): uttered by the complaining traveler, *five dollars* covers a price of \$4.90; uttered by the barista, it does not. The numeral stays the same but is read with more or less precision. The interpretations reflect which values count as five dollars in the current context: on the precise reading, only \$5.00 itself; on a loose one, a band of prices around it. I take these sets of values as basic objects.

Let D be the set of contextually possible values of the measured quantity: prices in the coffee case, counts of books in (3a), kilometers in (13a). Following Solt (2015), I assume a degree-based semantics for measure expressions: quantities are degrees of type d , and D is a set of such degrees. Whether these degrees are integers, reals, or points along some other dimension of measurement depends on the items being counted or measured. A **resolution description** for a numeral n is a set $h \subseteq D$ containing the values that count as n when the numeral is read at some

² For example, in approximated speech (e.g., *She said, approximately, ‘I am exhausted’*), the space of valid paraphrases (*tired, drained*) offers no total ordering: distinct alternatives may be equally close to the original. Neighborhoods in such domains are plausibly exemplar-based rather than scale-generated.

level of precision. Values are thus of type d , and a description, being a set of them, is of type r , which I use for resolution descriptions throughout. For (1), one can imagine $h_1 = \{\$5.00\}$ or $h_2 = \{\$4.90, \dots, \$5.10\}$.

A context typically makes several such readings available at once. I collect them into a family: $\mathcal{H}_n \subseteq 2^D$ is the family of resolution descriptions contextually available for n , where 2^D is the collection of all sets of values drawn from D . For a concrete case, take *I arrived at seven o'clock*. A plausible resolution description family for *seven* might be the following:

$$(20) \quad \mathcal{H}_7 = \{ \{7:00\}, \{6:59, 7:00, 7:01\}, \{6:55, \dots, 7:05\} \}$$

Which descriptions a context supplies is an empirical matter, and the uncertainty about the choice among them is put to work in Section 6; for now, we only need the family itself.

The resolution descriptions are not arbitrary. Let $\mathcal{H}_n$ be **centered** if every $h \in \mathcal{H}_n$ contains n , and assume that the families the context supplies are centered (Appendix A, Assumption 1). Intuitively, while a loose reading of *seven o'clock* might include values a little earlier or later than seven, it cannot exclude the value seven itself.

One member of a centered family plays a special role. Let E_n be an **exactness condition**: a predicate over descriptions that holds of h when reading n as h is reading it exactly. I write $F_n = \{h \in \mathcal{H}_n : E_n(h)\}$ for the exact descriptions in the family and assume F_n has at most one member, the **precision witness** p_n , so that $F_n = \{p_n\}$. For the family in (20), $p_7 = \{7:00\}$ —an exact reading of *seven o'clock* is one that admits only the value 7:00. Intuitively, this corresponds to the most precise reading of n possible; in terms of granularity scale-based approaches, F_n contains the interpretation of n under the finest level of granularity.

In a numeric domain such as the one presented, uniqueness is immediate, as $\{n\}$ is the only singleton containing n . I nonetheless state the assumption abstractly on purpose: the precision witness need not be the least description under inclusion, and in domains where descriptions are only partially ordered or not ordered at all, such as the paraphrase domains mentioned in Section 3.3, an exact reading need not be a minimal one.

The possible loose, approximate interpretations of n correspond to the non-exact resolution descriptions, which are derived by simply removing the exact descriptions from the family:

$$(21) \quad A_n = \mathcal{H}_n - F_n$$

For the running example, $A_7 = \{\{6:59, 7:00, 7:01\}, \{6:55, \dots, 7:05\}\}$.

In measurement domains, resolution descriptions can be generated on the basis of granularity scales presented in prior accounts. Let a **scale** g be a set of contextually salient points of D : values such as the multiples of five, ten, the quarter-hours of a clock, etc. Notably, scales need not be evenly spaced; an exponential scale, salient in domains such as computer memory, has points that spread further apart as they grow. Which values a salient point stands for depends on which points they are closer to, so I assume that values can be compared for closeness: for each value $v \in D$, a strict partial order $\prec_v$ on D , where $n \prec_v m$ means that n is closer to v than m is. The only constraint is that every value is closest to itself: $v \prec_v m$ for all $m \neq v$. For measurement domains, one natural way to compare closeness is by absolute difference: $n \prec_v m$ holds when $|v - n| < |v - m|$.

I refer to the description that a numeral n generates on a scale g as its **cell**: the set of values that round to n at that level of resolution:

$$(22) \quad \text{cell}(n)(g) = \{v \in D : n \prec_v m \text{ for every } m \in g \text{ such that } m \neq n\}$$

In words, a value v belongs to the cell of n on g if n is the salient point of g closest to v . With closeness compared by absolute difference, the cell's boundaries fall at the midpoints between n and its scale neighbors.³ For a scale with a fixed minimal unit γ , the definition then reduces to the symmetric interval $(n - \gamma/2, n + \gamma/2)$, matching prior accounts. For a non-uniform scale such as the exponential one, cells come out asymmetric.

A family of resolution descriptions then arises by letting n generate its cell on each contextually salient scale that contains n . For example, for $n = 47$, the finest scale, on which every value in D is salient, has 46 and 48 as the nearest neighbors of 47, and the cell is therefore the singleton $\{47\}$. A coarser scale keeps 47 salient but spreads its neighbors, widening the cell. Note that a family generated this way is centered by construction, since n is the closest salient point to itself on any scale that contains it. This corresponds to the requirement that granularity functions map each point to an interval containing it (requirement (6a) of Gyarmathy 2017), inherited by every cell. The finest scale, in turn, supplies the precision witness: in measurement domains, E_n holds of a cell just in case that cell is $\{n\}$, which is exactly what the finest scale generates.

4.2. NUMERALS AND RESOLUTIONS. Before turning to the approximating devices, we need to decide what a bare numeral contributes. I take as the baseline what Bylinina & Nouwen (2020) call the number view: a numeral denotes a number, here a degree of type d , and it enters the truth conditions through a counting operation in the spirit of Hackl (2000). On this view, *Roman bought seven books* is true when the number of books Roman bought equals seven (see also Kennedy 2015). I keep this composition and change one thing about it: the count is not compared to the number n but required to fall within some resolution description h of n , and the bare numeral leaves open which member of the family $\mathcal{H}_n$ that description is. In measurement domains, where resolution descriptions correspond to cells, this amounts to requiring the count to fall within the cell of n on some contextually salient scale.

An expression that leaves open which member of a set it involves is the signature of an indefinite. For example, *a cat sleeps* does not settle which cat sleeps; it is true as long as some cat does. I argue that bare numerals (and approximators more generally) exhibit quite similar properties. To model this parallel, I turn to the effect-driven interpretation framework of Bumford & Charlow (2026).

The framework starts from an observation about composition: alongside its ordinary contribution, an expression may carry a side computation such as an unresolved choice, a presupposition, or a dependency on context. Rather than building this into the ordinary meaning, Bumford & Charlow keep track of it separately, as an **effect**. Ordinary meanings then compose as usual, and the effects are threaded through the composition by a small set of general operations.

Indefinites carry the effect Bumford & Charlow call **Indeterminacy**, written S . The idea is that an indefinite does not denote one individual but keeps a set of candidates in play. Formally, an expression of type $\boxed{S\alpha}$ denotes a set of values of type α rather than a single one; these are the alternatives it leaves open.⁴ So *a cat* denotes the set of cats, $\llbracket a \text{ cat} \rrbracket = \{x : \text{cat}(x)\} : \boxed{Se}$.

³ Note that the inequality is strict, so a value exactly equidistant between two salient points falls in neither cell. This is in line with prior accounts such as Solt (2015) and Gyarmathy (2017).

⁴ Following Bumford & Charlow, I mark effect constructors with colored tiles and enclose an effectful type in a thin outline. The coloring has no semantic significance; it only shows at a glance how effects are distributed across a type.

Suppose there are two cats in the model, Almond and Biscuit; then $\llbracket a \text{ cat} \rrbracket = \{a, b\}$.

Composition now faces a mismatch: the predicate $\text{sleep} : e \rightarrow t$ expects a single individual, and the indefinite supplies a set of them. The framework resolves this without having to change the predicate, by positing that an ordinary function may be applied inside the effect wrapper, that is, to each alternative separately, collecting the results into a new set. Bumford & Charlow write this **map** operation as $\bullet_S$. Applying sleep across the alternatives gives a set of two truth values, one for each candidate:

$$(23) \quad \llbracket a \text{ cat sleeps} \rrbracket = \text{sleep} \bullet_S \{a, b\} = \{\text{sleep}(a), \text{sleep}(b)\} : (\mathbf{S}t)$$

The sentence *a cat sleeps* is then true if some candidate makes it true. This final step is existential closure, $\exists\text{-clo} : (\mathbf{S}t) \rightarrow t$, which returns truth if at least one alternative is true. Three operations, then, contribute to how the sentence is interpreted: the indeterminacy effect $\mathbf{S}$ supplies alternatives, $\text{map} \bullet_S$ applies the rest of the sentence to each of them, and $\exists\text{-clo}$ asks whether any of them succeeds.

Indeterminacy $\mathbf{S}$ is the effect-driven rendering of alternative semantics (Hamblin 1973; Rooth 1985; Charlow 2020), in which an expression contributes a set of alternatives that the rest of the sentence is evaluated against. My proposal is that what a numeral makes indeterminate is the resolution at which it is interpreted. The numeral supplies alternatives just as *a cat* does, and the alternatives are the descriptions in $\mathcal{H}_n$.

Writing r for the type of resolution descriptions, the bare numeral denotes its family:

$$(24) \quad \llbracket n \rrbracket = \mathcal{H}_n : (\mathbf{S}r)$$

The numeral does not denote the number n alone; it introduces the alternatives for how finely n may be read. Take (1), *a cup of coffee costs five dollars*, and for simplicity, assume prices are set in ten-cent steps, so that D contains \$4.80, \$4.90, \$5.00, and so on. A plausible family for *five* is the following:

$$(25) \quad \llbracket \text{five} \rrbracket = \mathcal{H}_5 = \{ \{5.00\}, \{4.90, 5.00, 5.10\}, \{4.80, 4.90, 5.00, 5.10, 5.20\} \}$$

The numeral thus introduces an indeterminate choice among the exact price, a loose reading to within ten cents, and a looser one to within twenty, parallel to the indeterminate choice among cats that *a cat* introduces.

The rest of the sentence supplies what to do with each of the alternatives introduced by the numeral. In *a cat sleeps*, it is the predicate sleep , which takes each candidate cat and returns the proposition that it sleeps. Here, the rest of the sentence takes each candidate description and returns the proposition that the actual price falls within it. This proposition is based on the actual quantity, which I denote as Q (here, what the coffee in fact costs), as well as the description h under consideration. I write the corresponding function on descriptions as $\text{scope}_Q : r \rightarrow t$, defined as follows:

$$(26) \quad \text{scope}_Q(h) = Q \in h$$

In words, scope_Q takes a description h and returns the proposition that the actual price Q falls within h . It is an ordinary function on descriptions, so it composes with the numeral exactly as sleep composed with *a cat*, by being mapped across the alternatives:

$$(27) \quad \text{scope}_Q \bullet_S \mathcal{H}_5 = \{ Q \in \{5.00\}, Q \in \{4.90, 5.00, 5.10\}, Q \in \{4.80, \dots, 5.20\} \} : \boxed{\mathbf{S} t}$$

The result is a set of three propositions, one for each description h . Existential closure disjoins them, returning true when Q falls within some available description:

$$(28) \quad \exists\text{-clo}(\text{scope}_Q \bullet_S \mathcal{H}_5) = Q \in \{5.00\} \vee Q \in \{4.90, 5.00, 5.10\} \vee Q \in \{4.80, \dots, 5.20\}$$

To see the whole derivation run, recall the traveler's context from Section 1, where a price of \$4.90 was enough for the sentence to count as true. Evaluating each disjunct at $Q = 4.90$:

$$(29) \quad \begin{array}{ll} 4.90 \in \{5.00\} & \text{false} \\ 4.90 \in \{4.90, 5.00, 5.10\} & \text{true} \\ 4.90 \in \{4.80, \dots, 5.20\} & \text{true} \end{array}$$

The exact description rejects the price, and both loose descriptions accept it; the sentence is true when read loosely. In a context where the resolution description family contains only the exact description, as it may in the case when the barista quotes the exact price, the disjunction reduces to $Q \in \{5.00\}$, and \$4.90 makes the sentence false. Which descriptions a context supplies, and how they are weighted where several are, is taken up in Section 6.

4.3. APPROXIMATORS AS ARTICLES OF PRECISION. The bare numeral leaves the choice of resolution open. The approximating devices, I now propose, are the expressions that manage this choice, and they do so in ways familiar from reference to individuals. *Approximately* narrows the alternatives without picking one, as an indefinite does; *exactly* picks out a unique one and fails if there is none, as a definite does. The material the devices operate on has been established: a numeral denotes the family of its candidate readings, the set of alternatives the device narrows or selects from. Devices such as *approximately* follow the indefinite pattern and return a set of descriptions; the effect is Indeterminacy, already in hand. Precision devices such as *exactly* follow the definite pattern instead, and definites need one more effect, Partiality, introduced in what follows.

I start with imprecise approximators such as *approximately*. Recall how *a cat* is formed: the noun *cat* denotes the predicate $\text{cat} : e \rightarrow t$, and the indefinite brings in an Indeterminacy effect $\mathbf{S}$ to return a set of things satisfying this predicate. Written out, the indefinite is a function from a predicate to a set:

$$(30) \quad \begin{array}{l} \llbracket a \rrbracket : (e \rightarrow t) \rightarrow \boxed{\mathbf{S} e} \\ \llbracket a \rrbracket(P) = \{x : P(x)\} \end{array}$$

It follows that $\llbracket a \rrbracket(\text{cat}) = \{x : \text{cat}(x)\}$, the set of cats.

I argue that *approximately* plays a similar role over resolution descriptions, with one difference in what it is handed. A common noun denotes a plain predicate, whereas a bare numeral already denotes its family of descriptions $\mathcal{H}_n : \boxed{\mathbf{S} r}$. So where an indefinite article a builds the alternatives from a predicate, *approximately* receives an already built set of alternatives and narrows them, returning the non-exact readings:

$$(31) \quad \begin{aligned} \llbracket \textit{approximately} \rrbracket : \boxed{\mathbf{S}r} &\rightarrow \boxed{\mathbf{S}r} \\ \llbracket \textit{approximately} \rrbracket (X) &= \{h \in X : \neg E_X(h)\} \end{aligned}$$

where E_X is the exactness condition of the family X .

The approximating device contributes its characteristic condition, that the description not be exact, by applying said condition to whatever family it is handed. In *approximately n*, that family, $\mathcal{H}_n : \boxed{\mathbf{S}r}$, is the denotation of the numeral.⁵ The two compose by ordinary function application, and the result is precisely A_n from Section 4.1:

$$(32) \quad \llbracket \textit{approximately } n \rrbracket = \{h \in \mathcal{H}_n : \neg E_n(h)\} = A_n : \boxed{\mathbf{S}r}$$

Where the bare numeral left every resolution description in $\mathcal{H}_n$ open, *approximately n* leaves open only the loose ones. For the coffee price example in (25), these are the two loose readings, with the singleton $\{5.00\}$ removed: $\{\{4.90, 5.00, 5.10\}, \{4.80, \dots, 5.20\}\}$. The device simply removes the exact reading; it does not build an interval, nor does it choose among the readings that remain. From here, composition runs as it did for the bare numeral. The rest of the sentence contributes scope_Q , mapping applies it to each remaining alternative, and closure disjoins:

$$(33) \quad \begin{aligned} \exists\text{-clo}(\text{scope}_Q \bullet_S A_5) &= \bigvee \{Q \in h : h \in A_5\} \\ &= Q \in \{4.90, 5.00, 5.10\} \vee Q \in \{4.80, \dots, 5.20\} \end{aligned}$$

Notably, the value n itself is not excluded, as it belongs to every remaining description by centeredness. An approximated numeral is thus interpreted at some non-exact resolution, with the choice among the remaining descriptions left to context.

I now turn to precision devices such as *exactly*. Where an indefinite keeps a set of candidates, a definite picks out one and requires that there be exactly one to pick: *the cat* refers to the unique cat if there is one and fails to refer otherwise. Bumford & Charlow model this with a second effect, **Partiality**, written $\mathbf{M}$. An expression of type $\boxed{\mathbf{M}\alpha}$ either returns a value of type α or fails. Failure, written $\#$, propagates through composition. A definite description is then $\llbracket \textit{the cat} \rrbracket : \boxed{\mathbf{M}e}$, returning the unique cat if the set of cats is a singleton and $\#$ if it is empty or has several members: a presupposition failure, handled as a computational effect.

Exactly, I argue, functions as the definite over resolutions. Taking a set of alternatives as an input, it returns the unique exact member of the family it receives, failing if there is none or more than one:

$$(34) \quad \begin{aligned} \llbracket \textit{exactly} \rrbracket : \boxed{\mathbf{S}r} &\rightarrow \boxed{\mathbf{M}r} \\ \llbracket \textit{exactly} \rrbracket (X) &= \begin{cases} h & \text{if } \{h' \in X : E_X(h')\} = \{h\} \\ \# & \text{otherwise} \end{cases} \end{aligned}$$

For *exactly n*, the family is again the numeral's denotation $\mathcal{H}_n : \boxed{\mathbf{S}r}$. Applying the definite-like *exactly* succeeds only if the set of satisfying descriptions is a singleton. This set is F_n , the set

⁵ The host need not be a (single) numeral: any expression that denotes a description anchors a family in the same way, with the description as its exact reading and coarser descriptions as its loose readings. For example, a range such as *15 to 20 days* denotes $\{15, \dots, 20\}$ with its bounds read exact; *approximately 15 to 20 days* denotes its loose readings, so that *21 days* satisfies the latter but not the former.

of exact descriptions. By the uniqueness assumption of Section 4.1, the only exact description of n is the precision witness p_n , so *exactly* retrieves it:

$$(35) \quad \llbracket \text{exactly } n \rrbracket = p_n : \boxed{\text{M } r}$$

Where *approximately* n left the loose readings open, *exactly* n settles on the exact one. The difference shows in the types: *approximately five* is a set of descriptions, of type $\boxed{\text{S } r}$; *exactly five* is one description, of type $\boxed{\text{M } r}$. For the coffee price family, that description is $p_5 = \{5.00\}$, the reading on which only the precise price of \$5.00 counts as five dollars.

Composition again applies the rest of the sentence inside the effect, now via $\bullet_{\text{M}}$: a function is applied to the value if there is one, and $\#$ is passed through if there is not. Since there is a value here, we get $\text{scope}_Q \bullet_{\text{M}} p_5 = Q \in \{5.00\}$, a single proposition, true at $Q = 5.00$ and false at $Q = 4.90$. There is no closure step: closure disjoins over alternatives, and *exactly five* carries none.

Notably, both entries can fail. *Exactly* fails when the family it receives has no exact member, so there is nothing to retrieve, and $\#$ is returned. This is presupposition failure. On the other hand, *approximately* fails when there is nothing left once the exact description is removed; that is, when the context supplies no non-exact description of n . The indefinite then returns the empty set of alternatives, $\emptyset : \boxed{\text{S } r}$; and closure disjoins over no alternatives, returning false. A failed *exactly* as such yields a sentence with no truth value, whereas a failed *approximately* yields a sentence that is false. Whether a context supplies exact/non-exact descriptions plausibly varies with the numeral and the domain: e.g., for small numerals in domains where things come in whole units, the loose readings might sit so close to the exact one that they may not be available at all.

4.4. APPROXIMATOR STACKING. With both an approximator and its precise counterpart in hand, I can now turn to what happens when such devices combine. Recall (4c), where three devices are stacked on a single numeral:

- (36) Roman kupyv prybyzno knyh zo sim.
 Roman bought approximately books-GEN.PL APPRX seven
 ‘Roman bought approximately seven books.’

As established earlier, (36) conveys the same interval as (3a) does with a single device. Before deriving this, I state the assumptions I am making about how such stacks are composed.

In the previous section, I provided the semantics for overt modifiers only. Aside from those, Ukrainian licenses indefinites, prepositional approximation, and approximative inversion; I suggest that each of these devices restricts the numeral’s family to its non-exact descriptions. I call this the **shared-condition assumption**.⁶ Focusing on the semantics, I remain agnostic about how each does so syntactically.

A second assumption concerns what the devices target. In a stack such as (36), both devices are hosted by the same numeral *sim*, and I take them to restrict the same family as conditions on

⁶ Treating all imprecise devices as restricting the resolution description family in the same way is a simplifying assumption. The framework does not depend on it: were the devices to impose different but compatible conditions, they would still stack felicitously, with the family restricted by both conditions at once, narrowing it rather than leaving it as either device alone would. Such flexibility may be needed for epistemic approximators, which I set aside here.

how *sim* is read. I call this the **shared-target assumption**. It is what distinguishes stacking from the discourse cases of Section 5, where an inner device targets a family already established by an earlier utterance and the shared-target assumption fails.

With the condition and the target both fixed, approximator stacking reduces to intersection. Each device, I argue, contributes a constraint on the resolution description family. A condition on resolution descriptions can be identified with the set of descriptions that satisfy it; the condition of being non-exact, for instance, is the set A_n . Restricting the family X by a condition R then means intersecting with it, and I write this as C_R :

$$(37) \quad C_R(X) = X \cap R$$

Approximately n returned the non-exact descriptions of n ; in this notation, that is $C_{A_n}(\mathcal{H}_n)$. *Exactly* n retrieved the exact one; that is the unique member of $C_{F_n}(\mathcal{H}_n)$. An imprecise device is thus C_{A_n} , and a precision device C_{F_n} .

When two devices apply in turn, the family is restricted twice. Restricting by R and then by T keeps the descriptions that satisfy both, which is the same as restricting once by $R \cap T$, and the order does not matter:

$$(38) \quad C_R \circ C_T = C_{R \cap T} = C_T \circ C_R$$

One immediate consequence of (38) is idempotence. Restricting by the same condition twice is restricting by it once, since $R \cap R = R$:

$$(39) \quad C_R \circ C_R = C_R$$

Under the shared-condition assumption, every imprecise device in a stack is C_{A_n} . A stack of any number of them therefore collapses to a single C_{A_n} , and applied to the family, it returns A_n , exactly as one device does. For the three-device stack in (36):

$$(40) \quad \llbracket \text{prybylżno knih zo sim} \rrbracket = C_{A_7}(C_{A_7}(C_{A_7}(\mathcal{H}_7))) = C_{A_7}(\mathcal{H}_7) = A_7 = \llbracket \text{prybylżno sim knih} \rrbracket$$

The stack and the single device denote the same set of alternatives, so composition with the rest of the sentence, and closure, proceed identically. The stacked forms thus result in the same interpretation as if only one approximator was used, and convey neither a wider nor a narrower interval than any single device.

Each imprecise device returns the type it takes, so a stack of them has the type of a single device, $\langle \mathbf{S} \, r \rangle$. A stack of two devices of differing precision is another matter. *Exactly* returns $\langle \mathbf{M} \, r \rangle$, which is not what *approximately* takes; and *approximately* returns a family with no exact member, which gives *exactly* nothing to retrieve. This is what I turn to next.

A second consequence of (38) is that two restrictions can exclude each other, as the intersection $R \cap T$ can be empty. An imprecise device is C_{A_n} and a precision device is C_{F_n} , and by construction $A_n \cap F_n = \emptyset$: the exact description is in the one and not the other. Under the shared-target assumption, both devices in a mixed stack restrict the same family, so the descriptions that satisfy both of the conditions are:

$$(41) \quad C_{F_n}(C_{A_n}(\mathcal{H}_n)) = C_{A_n}(C_{F_n}(\mathcal{H}_n)) = \mathcal{H}_n \cap A_n \cap F_n = \emptyset$$

No description is both exact and non-exact, so a mixed stack has nothing to denote. By the commutativity in (38), the order of the conditions does not matter. With *exactly* outermost, as in (16b), it can apply directly to $A_n : \boxed{\text{S } r}$, the denotation of *approximately n*. A_n has no exact member, and # is returned. In the case where *approximately* is outermost, as in (16a), it receives the denotation of *exactly n*, of type $\boxed{\text{M } r}$. A value of this type is either a description p_n or #, while *approximately* expects a family of type $\boxed{\text{S } r}$. Composition therefore has to get inside the Partiality layer and, if a description is there, hand it to *approximately* as a family. Turning a single description into a family can be done by the **unit** operation (cf. Bumford & Charlow 2026:(3.24)), which applies to any operator that, like *approximately*, takes a computation as its argument, and simply wraps the plain value into a trivial computation of the required effect. For Indeterminacy, this is the singleton: p_n turns into $\{p_n\} : \boxed{\text{S } r}$, a family with one member and no choice to make. Then, using $\bullet_M$, *approximately* can be applied to $\{p_n\}$ inside the M layer. Since the only member of that family is the exact description, the device removes it and nothing remains. The result has type $\boxed{\text{M } \text{S } r}$: if *exactly n* succeeded, an empty family by (41); if it failed, #, passed through.⁷

The infelicity of mixed stacks is a matter of content: two conditions that contradict on a shared target. As we have seen, however, such combinations are not uniformly bad. Because the contradiction depends on both devices targeting the same resolution family, the clash can be avoided if the inner approximated expression has already been established as a discourse referent. I detail the dynamic mechanics of this rescue in the next section.

5. Resolutions in discourse. Stacking a precise and an imprecise device on a single numeral leaves nothing to denote: the two devices target the same resolution family, and impose mutually exclusive conditions, so no description satisfies both. Asserted out of the blue, *exactly approximately n* and *approximately exactly n* are correctly predicted infelicitous, as seen in (16b) and (16a). However, such combinations are rescued when the inner approximated phrase is already established in the discourse (Zaroukian 2013). Solt (2015) observes the same pattern for comparatives, where *more than about 100* is fine as a reply that targets an earlier *about 100*.

I argue that this rescue is a matter of discourse anaphora. An approximated numeral, once asserted, introduces the resolution at which it is read as a discourse referent, and a later occurrence of the phrase can refer back to it rather than introduce a fresh one; the outer device then operates over that retrieved referent, in whichever order the two devices come. This breaks the shared-target assumption and avoids the clash. Modeling this requires that a resolution, once introduced, be available to later expressions, and hence a semantics in which the state of the discourse is carried from one utterance to the next. I adopt probabilistic dynamic semantics (PDS; Grove & White 2024) as that framework.

PDS provides the tools for lifting formal systems into a dynamic and probabilistic setting without altering the underlying compositional semantics. Because a range of formal theories can

⁷ Stacked precision devices such as *exactly at 5 o'clock on the dot* can be composed by the same mechanism. The outer device is applied to $\{p_n\}$ inside the Partiality layer, yielding a value of type $\boxed{\text{M } \text{M } r}$. Importing the **join** operation (cf. Bumford & Charlow 2026:(4.22)) allows flattening the two layers into one, so that the stack has type $\boxed{\text{M } r}$. Thus, semantically, the proposal treats stacks of precision devices as well-formed as those of imprecise approximators. I take the infelicity of English *exactly precisely* to be morphosyntactic, on par with the same-class restrictions of Section 2.2.2.

be implemented in it, including Rational Speech Act models of speaker-listener reasoning (Frank & Goodman 2012; Goodman & Lassiter 2015), adopting it does not by itself commit us to any one of them. The dynamic layer of PDS lets us model the changes in the discourse state, necessary for mixed-precision stacking rescue. The probabilistic layer, in which the alternatives an expression leaves open are weighted and marginalized over, is how I derive the graded, peaked interpretation of approximated numerals in Section 6.

5.1. INTRODUCING AND RETRIEVING RESOLUTIONS. In PDS, an utterance is interpreted against a **discourse state**, and asserting it yields an updated one. This state is formalized as a tuple of metalinguistic parameters, which typically includes the common ground (Stalnaker 1978), the Question Under Discussion (QUD) stack (Roberts 2012), and other contextually relevant information (Grove & White 2024). Grove & White deliberately take the state to be an open-ended collection of such parameters, so that it can be structured according to the needs of the particular model.

To model anaphoric rescue, I introduce an additional parameter to this tuple: the **referent store**, which serves as a persistent record of discourse referents available for subsequent anaphora. Just as the common ground tracks what is taken to be true and the QUD stack tracks what is under discussion, the referent store tracks what has been introduced and can be referred back to. My proposal is that resolutions are among the things it tracks. When an approximated numeral is asserted, the resolution description it was read at is added to the store, making it available for retrieval by later expressions.

The introduction and retrieval of resolutions can be modeled via two additional computational effects, both drawn from the framework of Bumford & Charlow (2026). An expression that makes a value available to later expressions carries the **Output** effect, written **W**. Formally, an expression of type $\boxed{\mathbf{W}}\alpha$ returns an ordinary value of type α together with a value it outputs (for our purposes, a resolution description), which is written to the store. Conversely, an expression whose interpretation depends on a previously established value carries the **Input** effect, written **R**. An expression of type $\boxed{\mathbf{R}}\alpha$ is a function awaiting a value, here again a resolution description, and returns a value of type α once it is supplied.

Output and Input are the effects Bumford & Charlow assign to antecedents and pronouns, respectively. Consider the sentence *Roman's cat scratched his hand*, in which the pronoun *his* refers back to *Roman*. The proper name introduces an individual, and alongside its ordinary contribution, it adds that individual to the referent store. Bumford & Charlow obtain this with the operator $\triangleright := \lambda x. \langle x, x \rangle$ of type $e \rightarrow \boxed{\mathbf{W}}e$, which pairs a value with itself as an output (their (5.1b)). Applied to *Roman*, it yields $\langle r, r \rangle$ of type $\boxed{\mathbf{W}}e$. The pronoun *his*, on the other hand, introduces no individual of its own; it awaits one, and its semantic value is the identity function $\text{id} = \lambda x. x$ of type $\boxed{\mathbf{R}}e$, which returns the referent it receives.

When the two compose, the written referent r and the pronoun's request id are both carried along, and the sentence ends up as a request paired with a referent, $\langle \text{id}, r \rangle$. Discharging the pair is the operation Bumford & Charlow call the **co-unit**, which applies the request to the supplied value:

$$(42) \quad \varepsilon(\langle \text{id}, r \rangle) = \text{id}(r) = r$$

The pronoun resolves to *Roman* (cf. Bumford & Charlow 2026:(5.9)). Output and Input effects stand in a systematic correspondence—formally, an **adjunction**. An expression that sup-

plies a value and an expression that requests one are two sides of a single dependency; the co-unit operation is the mechanism that discharges one instance of this dependency.

Within a sentence, an Output and an Input effect meet by composition, as in the example above. Across utterances they do not, and this is where the referent store comes in. Asserting an utterance appends the result of its Output effects to the store, which is carried forward as part of the discourse state. An Input effect that reaches the utterance boundary is discharged against the referent store: its request is paired with the corresponding stored value and the co-unit applies, exactly as within a sentence. Within PDS, the store is one more parameter of the state that assertion updates and later utterances read.

My claim is that anaphora to a numerical resolution operates through the same mechanism: approximated numerals, like proper names, are referent-introducing. *Approximately* n restricts the family to A_n and leaves the choice among the loose descriptions open, and *exactly* n retrieves p_n . In addition, like the proper name in the example above, the referent store is updated, and by the same operator: $\triangleright$ applies to descriptions as it applies to individuals, pairing a description with itself as an output.

Since *approximately* n leaves several descriptions open, the output is made inside the Indeterminacy effect, each alternative outputting itself, so that whichever description is in play is the one recorded:

$$(43) \quad \triangleright \bullet_S A_n = \{ \langle h, h \rangle : h \in A_n \} : \boxed{\text{S W } r}$$

With *exactly* n , the write takes place inside the Partiality effect. This ensures that a referent is stored only if retrieval succeeded: if *exactly* n is $\#$, $\bullet_M$ passes the failure through and nothing is stored:

$$(44) \quad \triangleright \bullet_M p_n = \langle p_n, p_n \rangle : \boxed{\text{M W } r}$$

In both cases, the store records a description $h : r$. A later occurrence of the numerical phrase has, besides its fresh reading, an anaphoric one, on which it functions like a pronoun. It introduces no resolution family of its own, but instead awaits a description. Its value is the identity function $\text{id} = \lambda h. h : \boxed{\text{R } r}$. Discharging the request against the store pairs it with the stored description, $\langle \text{id}, h \rangle$, and the co-unit resolves it:

$$(45) \quad \varepsilon(\langle \text{id}, h \rangle) = \text{id}(h) = h$$

The anaphoric reading is available only when a co-unit can discharge it, that is, only when an earlier expression has output a description.

Once retrieved, the resolution description is then **re-anchored**. Recall that a family collects the readings available for some description: e.g., $\mathcal{H}_n$ collects those available for $\{n\}$. A retrieved description, I argue, can anchor a family in the same way. Consider one of the descriptions an earlier *approximately* *five* leaves open, $h_{\text{ref}} = \{4.90, 5.00, 5.10\}$; by (43), each such description is written to the referent store. In the corresponding family $\mathcal{H}_{h_{\text{ref}}}$, the finer reading $\{5.00\}$ is gone, while h_{ref} , which was a loose reading of *five*, is now the exact reading $p_{h_{\text{ref}}}$. Re-anchoring thus changes what counts as exact: a resolution an earlier utterance established can itself be read exactly or loosely.

In general, for a retrieved description h_{ref} , $\mathcal{H}_{h_{\text{ref}}}$ is the family of readings available for h_{ref} , with $p_{h_{\text{ref}}} = h_{\text{ref}}$. Re-anchoring is the map from a description to its family, applied inside the Input effect by $\bullet_R$:

$$(46) \quad (\lambda x. \mathcal{H}_x) \bullet_R \text{id} = \lambda x. \mathcal{H}_x : \boxed{\text{R S } r}$$

I argue that this re-anchoring step is what the rescue of mixed-precision stacks relies on. An outer device applied to the anaphoric phrase operates on $\mathcal{H}_{h_{\text{ref}}}$, the family of an already-fixed resolution, and not on $\mathcal{H}_n$, the family the antecedent restricted. The two families are distinct targets, and conditions on them no longer clash.

5.2. THE MIXED-PRECISION RESCUE. Both the infelicity and the rescue of the mixed-precision cases can now be derived. Consider *exactly approximately n* in these two distinct contexts.

Out of the blue, the referent store contains no matching antecedent for the inner expression. Consequently, *approximately n* cannot refer back to anything and is computed fresh: it restricts the resolution family $\mathcal{H}_n$ to the non-exact subset A_n , leaving the choice among the loose descriptions open. The outer device *exactly* then targets the same family $\mathcal{H}_n$, and imposes its own condition that the resolution description be exact. Both devices thus condition the same family, and no description satisfies both. *Exactly* receives A_n , finds no exact member to retrieve, and returns $\#$, resulting in the out-of-the-blue clash observed in (16b).

Now suppose that *approximately n* has been asserted earlier in the discourse. By (43), that assertion wrote each loose description $h \in A_n$ it left open to the referent store. When the inner phrase of *exactly approximately n* is now interpreted, it has an antecedent and takes its anaphoric reading: it awaits a description, and re-anchoring by (46) gives $\lambda x. \mathcal{H}_x : \boxed{\text{R S } r}$. *Exactly* is applied inside the Input effect. The request is then discharged against the store, which pairs it with the stored description h_{ref} , and the co-unit collapses the pair:

$$(47) \quad \begin{aligned} \llbracket \text{exactly} \rrbracket \bullet_R (\lambda x. \mathcal{H}_x) &= \lambda x. \llbracket \text{exactly} \rrbracket (\mathcal{H}_x) : \boxed{\text{R M } r} \\ \in (\langle \lambda x. \llbracket \text{exactly} \rrbracket (\mathcal{H}_x), h_{\text{ref}} \rangle) &= \llbracket \text{exactly} \rrbracket (\mathcal{H}_{h_{\text{ref}}}) = p_{h_{\text{ref}}} = h_{\text{ref}} : \boxed{\text{M } r} \end{aligned}$$

Exactly thus applies to $\mathcal{H}_{h_{\text{ref}}}$, not to $\mathcal{H}_n$, and retrieves $p_{h_{\text{ref}}}$, which is h_{ref} itself. The resulting interpretation carries two compatible conditions: first, that the antecedent numeral was read at a non-exact resolution ($h_{\text{ref}} \in A_n$), and second, that this specific introduced resolution is now being targeted precisely ($p_{h_{\text{ref}}} \in F_{h_{\text{ref}}}$). These concern different families, $\mathcal{H}_n$ and $\mathcal{H}_{h_{\text{ref}}}$, and so do not conflict. The phrase is felicitous, and its meaning is “exactly that approximate value”: the speaker takes the loose reading of n that is already on the table and locates the quantity at it precisely.

The reverse ordering, *approximately exactly n*, follows the same logic. Out of the blue, *exactly n* yields the single description p_n , and *approximately* has nothing loose to keep: it removes the exact reading from $\{p_n\}$ and returns an empty family. Given an antecedent, the inner *exactly n* is anaphoric instead. It retrieves the stored description h_{ref} and re-anchors it to $\mathcal{H}_{h_{\text{ref}}}$, which has loose readings of its own. *Approximately* now has something to keep, and returns $A_{h_{\text{ref}}}$: the loose readings of the exact value already on the table.

This derives what Zaroukian (2013) describes as the prototype reading. On Zaroukian’s account, expressions like *approximately n* denote a prototype for the notion of being approximately n , and the outer device is taken to relate the quantity to this ideal; the reading requires a coerced

type-shift on the inner phrase. On ours, no shift is needed. The stored referent is an ordinary description, of the same type as any resolution, and the outer device applies to the family it anchors exactly as it applies to a numeral’s own family. The same mechanism accounts for Solt’s observation regarding comparatives: phrases such as *more than about 100* are ill-formed out of the blue, but acceptable in a reply that refutes a prior claim about *about 100*. On my account, the comparative requires *about 100* to be available as a discourse referent: with an earlier assertion of *about 100* in the store, the phrase retrieves the loose description of 100 that was introduced, re-anchors it, and feeds it to *more than*.

6. Probability and its consequences. As is, the proposal presented in prior sections is crisp: it derives the truth conditions of an approximated numeral for any fixed choice of resolution, but leaves the choice itself unresolved. I now introduce **uncertainty** into the picture. The grammar alone cannot select a specific description; as the coffee price example demonstrates, the intended level of precision is underdetermined by the sentence itself. Imprecision, on my account, is not an interval an approximator builds around a numeral; it is uncertainty about which of the resolution descriptions the numeral itself introduces is in play. I argue that the choice among descriptions is weighted, and the interpretation of a numeral becomes a distribution over quantities, obtained by averaging over the weighted descriptions. The peaked shape of approximated numerals, the contrast between *around* and *between*, and the roundness and magnitude phenomena all follow from this move.

6.1. LIFTING THE SEMANTICS. The baseline proposal leaves open, at every numeral, which specific resolution description it is actually read at. The Indeterminacy effect carries a set of alternatives; however, nothing is said about which of them is more likely. What is missing is a way of saying that some of those candidates are more likely than others. I argue that the compositional system does not need to be changed drastically to accommodate this fact; instead, we can use probability distributions where we previously used sets. The semantics computes the same values as before, but weighted, and the uncertainty over which interpretation is at play comes from the spread of those probabilities.

Viewing interpretation probabilistically is a general strategy, familiar from probabilistic semantics and pragmatics (Goodman & Lassiter 2015), where it is used to explain graded interpretation without altering the underlying semantics. Krifka (2009) also assigns scales prior probabilities and updates them on hearing a measure expression, deriving a preference for coarser scales. Hénot-Mortier et al. (2026) likewise model the peakedness of *around*-expressions probabilistically. My approach differs in two respects. First, where the Coarsest Scale Principle of Krifka selects a single winning scale, I retain the full distribution over resolution descriptions and marginalize over it, so that the peaked interpretation falls out of the structure of the family directly. Second, I locate this probability in the compositional semantics, modeling the uncertainty over the resolution in play within the common ground itself.

To do this, I bring in the probabilistic layer of PDS: the alternatives an expression leaves open are weighted and marginalized over, and assertion conditions the common ground on what was said.⁸

⁸ I do not model a speaker choosing a numeral or a listener reasoning about that choice, though the modularity of PDS means that a speaker model could be built alongside this account to supply the distributions I take as primitive.

6.2. THE PROBABILITY EFFECT. Lifting the approximator semantics described before amounts to a single substitution. Where the baseline system relied on the Indeterminacy effect S, carrying a flat set of alternatives, I now introduce a **Probability** effect P. While an expression of type $\boxed{S\alpha}$ denotes a set of alternatives of type α , an expression of type $\boxed{P\alpha}$ denotes a probability distribution over them, assigning each alternative a probability such that the total sums to one.

The compositional operations defined for S translate directly to counterparts in P. The map operation $(\bullet_P)$ still applies an ordinary function to each alternative, but now carries its associated probability along. Restriction by a condition, which under S discarded alternatives that failed to satisfy it, becomes **conditioning**: alternatives failing the condition have their probability set to zero, and the remaining weights are renormalized to sum to one. For a distribution θ over descriptions and a condition R , the restriction operation is thus defined as follows:

$$(48) \quad C_R(\theta)(h) = \frac{\theta_h \cdot \mathbb{1}[h \in R]}{\sum_{h' \in R} \theta_{h'}}$$

Finally, existential closure, which under S asked whether some alternative was true, becomes **marginalization**. It sums the weights of the alternatives that make the proposition true, returning not a truth value, but the probability that the proposition holds.

The lexical entries carry over under the substitution, with $\boxed{Pr}$ in place of $\boxed{Sr}$ in their types. The bare numeral introduces its resolution family, now with contextual weights attached; *approximately* keeps the non-exact descriptions, conditions a distribution on them; and *exactly* retrieves the unique exact description, now from the support of the distribution, and fails if there is none.

The facts about approximator stacking also carry over under the P effect. Conditioning twice on the same set is conditioning on it once, since the second pass finds no further probability to remove, so same-precision stacking is idempotent in P as it was in S. Conversely, conditioning on a set that contains no alternative is undefined, since the denominator in (48) becomes zero. The two orders of mixed-precision stacking from Section 2.2 carry over accordingly. With *exactly* outermost, it receives a distribution whose support contains no exact description and returns $\#$, as before. With *approximately* outermost, it conditions on the non-exact descriptions inside the M layer, and since the only description present is the exact one, the denominator in (48) is zero and the result is undefined. Both orders remain infelicitous out of the blue.

The rescue of Section 5 carries over as well: S is replaced with P; the Output W, Input R effects, and the co-unit are left untouched. (43) becomes $\triangleright \bullet_P C_{A_n}(\theta)$, each description written with its weight, and (47) does not change.

It remains to specify the source of the probabilities that now come attached to the resolution description alternatives. In PDS, the common ground is modeled as a probability distribution over indices, each of which pairs a world with a context (Grove & White 2024). To assert a sentence is to update the common ground (Stalnaker 1978). I argue that the interpretation of a numeral depends on two parameters: the actual quantity (fixed by the world), and how finely the numeral is read (fixed by the context). Both vary across the common ground, representing the uncertainty over how many things are actually in the world and the uncertainty over how finely they are counted, respectively.

I assign a prior to each. The resolution at an index is distributed according to a prior θ over the contextually supplied family $\mathcal{H}_n$, with θ_h representing the probability assigned to the description h . Which descriptions the family contains, and so what θ ranges over, is information the con-

text supplies. Similarly, the actual quantity Q at an index is distributed according to a prior ρ , which encodes background expectations about the quantity in question. I leave the exact form of θ and ρ open, and take the two priors to be independent.

The lifted entries can now be stated; take example (25): *a cup of coffee costs five dollars*. Suppose the context supplies the prior in (49), placing most of its mass on the exact reading:

$$(49) \quad \theta = \{ \{5.00\} \mapsto 0.6, \{4.90, 5.00, 5.10\} \mapsto 0.3, \{4.80, \dots, 5.20\} \mapsto 0.1 \}$$

Under S , the bare numeral denoted its family; under P , it denotes this prior:

$$(50) \quad \llbracket \text{five} \rrbracket = \theta : \boxed{\text{P}} r$$

Approximately five, under S , removed the exact descriptions from $\mathcal{H}_5$ and returned the set of loose ones, A_5 . The lifted device now does the same to the prior: it conditions θ on A_5 by (48), setting the exact description's probability to zero and renormalizing the rest:

$$(51) \quad C_{A_5}(\theta) = \{ \{4.90, 5.00, 5.10\} \mapsto 0.75, \{4.80, \dots, 5.20\} \mapsto 0.25 \}$$

The alternatives are the same as before, but each now carries a probability. Note that these local lexical entries rely entirely on θ , the prior over resolutions. The actual quantity prior, ρ , enters the derivation only when this probabilistic semantic value composes with the rest of the sentence, where the uncertainty over the resolution and the uncertainty over the world state are jointly evaluated.

On the other hand, *exactly* is affected by the substitution only in its input type, which becomes $\boxed{\text{P}} r$. The weights are not used, however; as before, *exactly* looks at which descriptions are present (now the support of θ) and retrieves the unique exact one among them, or fails. The output type is $\boxed{\text{M}} r$ as before, untouched by the lift. Its interpretation in context thus involves the world prior ρ but not the resolution prior θ : a definite retrieves its description rather than weighting alternatives, so there is no resolution uncertainty for θ to enter.

6.3. THE INTERPRETATION. Given the two priors and an asserted sentence with a numerical expression, I model the interpretation of that sentence as computing the probability of an answer to a question of interest. As stated above, asserting the sentence conditions the common ground on its proposition. To read off what the assertion conveys, one has to query the updated common ground.

Following Grove & White (2025a,b), I assume a function likely that takes a common ground and a proposition, returning the probability of that proposition by summing the probability mass of every index on which it holds. Writing CG for the common ground before an assertion and CG' for the common ground after it, the specific query we put to CG' is *How likely is it that Q takes a particular value m ?*, where Q is the actual quantity and m is some specific fixed value; that is, I compute likely(CG')($Q = m$). While PDS formally states both the update and the query over discourse states that thread the common ground between utterances, for simplicity I abstract away here from the discourse state and work with the common ground alone.

Consider first a bare numeral n , with no approximating device. The rest of the sentence contributes the proposition $\text{scope}_Q(h) = Q \in h$: that the actual quantity falls within the description h at which n is read. Asserting the numeral conditions the common ground on this proposition.

A bare numeral contributes no condition on the resolution beyond membership, so θ enters at its full prior: every description in the family remains in play, and the assertion is that the quantity falls into whichever one the index fixes. Before the assertion, an index on which the quantity is m and the description is h has prior probability $\rho(m) \cdot \theta_h$, the two priors being independent. The assertion keeps only the indices on which the quantity falls within the description, so an index carrying the pair $\langle m, h \rangle$ survives with probability proportional to

$$(52) \quad \rho(m) \cdot \theta_h \cdot \mathbb{1}[m \in h]$$

The indicator is what the assertion contributes: it is 1 when the quantity falls within the description and 0 otherwise. A quantity m that is far from n thus survives only under resolution descriptions h wide enough to reach it. To see this at work, take the coffee price example again, with the family $\mathcal{H}_5$ and the prior θ of (49), and assert *five dollars*. An index with $m = 5.00$ survives on every resolution, since 5.00 belongs to all three; an index with $m = 4.90$ survives only on the two loose descriptions; an index with $m = 4.70$ survives on none.

Having asserted the numeral, we now can ask what it conveys about the actual quantity. This is where the query comes in: across the updated common ground, how likely it is that the quantity takes a particular value m . Note that this is a different proposition from the one asserted: the assertion says that the quantity lies within some resolution description h of n , whichever it is, while the query asks whether the quantity equals a specific m .

Answering the query means summing (52) over every index on which $Q = m$. The quantity is fixed at m , leaving the single factor $\rho(m)$; the description, unconstrained by the query, is summed over. The result is then the following:

$$(53) \quad \text{likely}(\text{CG}')(Q = m) \propto \rho(m) \cdot \sum_{h \in \mathcal{H}_n} \theta_h \cdot \mathbb{1}[m \in h]$$

For the coffee price family, the sum in (53) is easy to read off. At $m = 5.00$, every description contains the value, and the sum is $\theta_{\{5.00\}} + \theta_{\{4.90, 5.00, 5.10\}} + \theta_{\{4.80, \dots, 5.20\}} = 0.6 + 0.3 + 0.1 = 1$. At $m = 4.90$, only the two loose descriptions contain it, and the sum is $0.3 + 0.1 = 0.4$. At $m = 4.80$, only the widest does, and the sum is 0.1. At $m = 4.70$, none does, and the sum is 0. The semantic factor is thus 1 at the exact value of the numeral, and drops in steps as we move away from it, each step falling at the edge of a description.

The interpretation of a numeral at a candidate quantity m is therefore proportional to a product of two factors. The first, $\rho(m)$, is the contribution of world knowledge, the prior expectation that the quantity is m . The second comes from the semantics of the numeral: it is the probability, marginalized over how finely the numeral is read, that m falls within a description of n . This product form parallels the literal listener update in probabilistic pragmatics, where the posterior over states is proportional to a prior over states times the truth of the utterance at each state (Goodman & Lassiter 2015). In the presented framework, the second factor is a probability rather than a truth value, since whether m verifies the numeral depends on the resolution.

6.4. DERIVING PEAKEDNESS. Recall the observation from Section 2.3.2: an approximated numeral does not convey a flat interval (Égré et al. 2023; Hénot-Mortier et al. 2026). *Around 70 people* suggests that values close to 70 are more likely to represent the actual quantity than values farther from it. By contrast, *between 60 and 80 people* treats every value in its range alike. The interpretation of an approximated numeral is, in this sense, peaked, with the peak anchored at the

numeral itself. On the presented account, this distinction can be derived from the semantic factor in (53).

The semantic factor, $\sum_{h \in \mathcal{H}_n} \theta_h \cdot \mathbb{1}[m \in h]$, sums the probability of every description that contains the candidate quantity m . Its shape follows from the structure of the family. Because the family is centered, n belongs to every available description. Thus, at $m = n$, every term in the sum is included, and the factor attains its maximum value of 1 (Appendix A, Proposition 1).

This is what places the peak of an approximated numeral’s interpretation at the numeral itself. It depends purely on membership: the value n is in every resolution description. Away from n , the factor is non-increasing: a quantity farther from n belongs to no more descriptions than a closer one because descriptions are solid, gapless stretches containing n . Thus, the factor can only stay level or drop as m moves outward (Appendix A, Theorem 1). Between the boundaries of any two descriptions, the factor is flat. In particular, if the narrowest available description is wider than $\{n\}$, the top of the peak forms a flat plateau spanning that description. Note that this fixes the location of the maximum, not of the mean or the median: if the available descriptions extend farther on one side of n than on the other, as with the one-sided descriptions hinted at in Section 4.1, the interpretation is skewed: most probable at n , but with more of its mass on the longer side.

The shape of the peak, within these constraints, is determined by the resolution prior θ . The distribution becomes sharper when θ concentrates its probability mass on narrow descriptions, causing the semantic factor to drop steeply just outside n . Conversely, it broadens when this mass is spread across wider descriptions. When the world prior ρ is uninformative, the final interpretation follows the semantic factor, yielding this bare peaked shape. When ρ is informative, however, it modulates the peak. Intuitively, this models how approximation is defeasible by knowledge of the facts.

I now turn to the contribution of approximators. Because *approximately* n conditions the resolution prior θ on non-exactness, as in (51), the exact description p_n no longer contributes to the sum, and the remaining weights are renormalized. The semantic factor for the approximated numeral thus becomes:

$$(54) \quad \sum_{h \in \mathcal{H}_n} C_{A_n}(\theta)(h) \cdot \mathbb{1}[m \in h]$$

Since $p_n = \{n\}$, the exact description contributes only at $m = n$, adding θ_{p_n} there and nowhere else. The bare numeral, on the other hand, places an extra spike of probability on the quantity being exactly n , which the approximator removes. Both interpretations remain peaked and centered on n . The difference is that the bare numeral places additional weight on the exact value, while *approximately* n does not, keeping n at the center of the peak while excluding the precise reading.

Returning to the running coffee price example, the difference becomes visible once we move off the numeral. For the bare numeral, the semantic factor is 1 at 5.00, 0.4 at 4.90, and 0.1 at 4.80. It drops sharply at the first step, because 0.6 of its prior mass rested on the precision witness, which contributes nowhere but at 5.00. By contrast, the approximated numeral’s factor, computed over θ renormalized to A_5 , is 1 at 5.00, 1 at 4.90, and 0.25 at 4.80. With the exact reading conditioned away, the top of the peak forms a flat plateau across the narrowest loose description, $\{4.90, 5.00, 5.10\}$, and only begins to drop beyond it. The bare numeral thus says that the price is \$5.00, and if not, then near it; *approximately five* says that it is somewhere in the cell around

\$5.00, without favoring the exact value. This contributes to the intuition the bare numeral’s default reading is the exact one.

The contrast with *between*-expressions is similarly derived. An explicit bound such as *between 60 and 80* denotes its interval directly, imposing a crisp truth-conditional constraint on the quantity itself: $Q \in [60, 80]$. Conditioning the common ground on this proposition and querying a quantity m as in (53) yields a posterior proportional to $\rho(m) \cdot \mathbb{1}[m \in [60, 80]]$: the prior truncated to the interval, uniform within it whenever ρ is. The contrast between the peakedness of *around* and the uniformity of *between* thus reduces to whether the interval is supplied by the expression or whether it emerges from marginalizing over resolution uncertainty.

Peakedness also requires that more than one resolution be available. If the context supplies a single description for n , there is no uncertainty to marginalize over; the semantic factor is an indicator, and *around n* coincides with *between*.

6.5. ROUNDNESS AND MAGNITUDE. I now return to the two empirical observations from Section 2.3.1, which have to do with roundness and magnitude. Round numerals such as *one hundred* invite a wider approximate interpretation than non-round numerals such as *ninety-eight*. This phenomenon persists under an approximator: *around one hundred* covers a broader range of values than *around ninety-eight*. Similarly, magnitude independently dictates width: larger numerals invite wider readings than smaller ones, such that *about 800* spans a larger absolute interval than *about 100* (Lefort et al. 2016, 2017). Both phenomena reflect the numeral’s own effect on the interval, independent of the approximator, and both can be located in the resolution prior θ .

While θ has so far been treated as fixed, it is naturally sensitive to the numeral. Which descriptions populate the family $\mathcal{H}_n$, and how much prior probability each carries, varies from one numeral to the next. Where a family is generated from scales, as in measurement domains, this is because a numeral is salient on some scales but not others. A specific coarse description is only available for n if n itself serves as a salient point on the corresponding scale.⁹

Similar to prior accounts (Krifka 2009; Sauerland & Stateva 2011; Solt 2015), I argue that round numerals license a richer and wider array of available descriptions than non-round ones. One could imagine a round numeral such as 100 as a salient point on the by-fives, by-tens, by-twenty-fives, and by-fifties scales, to name a few, generating a corresponding resolution description at each of these widths. By contrast, a non-round numeral such as 97 aligns with far fewer scales. Since the prior θ ranges only over the numeral’s family $\mathcal{H}_n$, a round numeral’s θ distributes probability mass across a range of wide descriptions, while a non-round numeral’s θ is concentrated on narrow ones (Appendix A, Assumption 3).

This difference reshapes the semantic factor accordingly. Because the probability mass for 100 is distributed across wider descriptions, the resulting peak is correspondingly broad: candidate quantities far from 100 still fall within the boundaries of a coarse resolution. Conversely, for 98, where prior probability is concentrated on narrow descriptions, the peak is sharper: only

⁹ Locating roundness in the prior over resolutions is one way to capture this phenomenon. An alternative approach derives it through speaker-listener reasoning: a round number is more likely to have been chosen as an estimate, so hearing one is evidence that a coarse resolution was intended (see, for example, Kao et al. 2014 and Waldon 2022). The presented account is compatible with such a model: the semantics requires some θ , and a model of the speaker is one thing that could supply it, in which case the roundness effect would enter θ via pragmatic inference over the speaker’s choices. For the present purposes, I treat θ as a given contextual parameter and leave its source to future work.

quantities close to 98 satisfy the available resolutions. The model thus predicts that *around one hundred* admits a wider spread of quantities than *around ninety-eight*, matching the contrast in (18): one may answer *around one hundred* when the distance is anywhere in a broad interval, whereas *around ninety-eight* invites a narrower one. The roundness phenomenon is, on this account, a fact about which descriptions a numeral licenses in a given context, and hence about the support of the resolution prior θ .

The magnitude pattern arises in the same way. For a numeral of larger magnitude, the coarse descriptions available in a given context are wider. A quantity like 800, for instance, is naturally reported on scales counting by hundreds or even two-hundreds, whereas the coarse scales invoked for 100 rarely extend to such widths. The prior θ for a larger numeral therefore distributes its probability mass across wider descriptions, inducing a correspondingly broader peak (Appendix A, Assumption 4 and Theorem 1). This predicts that *about 800* spans a wider interval than *about 100* (Lefort et al. 2016, 2017). Like roundness, it requires no additional mechanism. Both phenomena reduce to the fact that θ depends on the numeral, so that numerals induce broader peaks when the resolution descriptions that are available are wider. The contributions of the numeral and the approximator that Ferson et al. (2015) find predictive of interval width are the numeral’s effect on θ and the approximator’s condition on the same sum.

One further consequence of locating imprecision in a prior over resolutions is that the interpretation is not fixed across speakers. The peak is shaped by θ , and, where world knowledge bears on it, by ρ ; neither need be shared. Two speakers who differ in which resolutions they treat as contextually available, or in how they allocate probability mass across coarser resolutions, will read the same approximated numeral with differently shaped peaks, even holding the numeral and the approximator constant. The probabilistic layer captures this directly: inter-speaker variation in the interpretation of approximation is variation in θ and ρ .

7. Conclusion. I have argued that approximators neither widen a numeral’s denotation nor set a granularity parameter. Instead, they interact with resolutions in the way that indefinites and definites interact with individuals. A bare numeral introduces a family of resolution descriptions and leaves the choice among them open. Imprecise devices such as *approximately* narrow this choice to the non-exact descriptions without settling it, as an indefinite does; precision devices such as *exactly* retrieve the unique exact description and presuppose that there is one, mirroring a definite. Because both devices contribute conditions on the same family, stacking them amounts to set intersection. Same-precision stacks are idempotent; conversely, mixed-precision stacks impose mutually exclusive conditions. Resolving an approximated phrase either afresh or against a referent an earlier utterance introduced explains the out-of-the-blue clash of mixed-precision stacking and its rescue.

Lifting these alternatives to probability distributions, as probabilistic dynamic semantics allows, derives the remaining phenomena. Throughout, an approximator contributes only a condition on how its numeral is read; the phenomena follow from which family that condition targets and how the resolution descriptions in the family are probabilistically weighted.

Several questions I have raised remain open. First, I characterized the heightened hedging that stacked devices convey as pragmatic, arising from the form of the utterance rather than from its content, but have not formally modeled it. Second, I have taken the same-class restriction on stacking, which blocks *pyblyzno orijentovno* and English *approximately around* alike, to be morphosyntactic, and left its analysis aside. Third, Solt (2015) observes that approximated expres-

sions behave akin to negative polarity items under strict comparatives. This polarity sensitivity does not immediately fall out from what I proposed. Finally, approximators apply to expressions other than numerals; whether the present account extends to approximated quotation or categorization, where the descriptions would be exemplar-based rather than scale-generated, is a question I have only gestured at.

A. Proofs. In this appendix, I collect the formal assumptions on resolution families that the results of Sections 4.1 and 6 rely on, and provide the corresponding proofs.

Throughout, D is the set of contextually possible values. The results below are stated for the numeric instance of Section 4.1: D is linearly ordered, as is standard for quantities like prices, and closeness is compared by absolute difference, so that $n \prec_v m$ holds just in case $|v - n| < |v - m|$. A description is **convex** if, whenever it contains two values, it contains every value of D between them. For a numeral n with a resolution family $\mathcal{H}_n$ and a prior θ over it, the **semantic factor** presented in Section 6.3 is defined as follows:

$$(55) \quad S_n(m) = \sum_{h \in \mathcal{H}_n} \theta_h \cdot \mathbb{1}[m \in h],$$

Assumption 1 (Centeredness). For every $h \in \mathcal{H}_n$, $n \in h$. A resolution of *five* on which \$5.00 does not count as five dollars is not a resolution of *five*.

Assumption 2 (Solidity). Every $h \in \mathcal{H}_n$ is convex relative to D : for all $v_1, v_2 \in h$ and $u \in D$, if $v_1 \leq u \leq v_2$ then $u \in h$. A resolution description is thus a stretch of D with no gaps.

For scale-generated families, Assumption 2 is a consequence of the construction:

Lemma 1 (Cells are solid). For any scale g containing n , $\text{cell}(n)(g)$ is convex relative to D .

Proof. By (22), $v \in \text{cell}(n)(g)$ if and only if $|v - n| < |v - p|$ for every salient $p \in g$ with $p \neq n$. Let $v_1, v_2 \in \text{cell}(n)(g)$ with $v_1 < v_2$, and let $u \in D$ with $v_1 < u < v_2$. Suppose $u \notin \text{cell}(n)(g)$; then there is a salient $p \neq n$ with $|u - p| \leq |u - n|$. If $p > u$, then v_2 lies on the same side of u as p , and so is at least as close to p as u is, relative to n : $|v_2 - p| \leq |v_2 - n|$. Then $v_2 \notin \text{cell}(n)(g)$, contradicting the assumption. If $p < u$, the same argument applies to v_1 . Hence $u \in \text{cell}(n)(g)$. $\square$

Proposition 1 (Mode location). Under the centeredness assumption, $S_n(n) = 1 \geq S_n(m)$ for all $m \in D$. The numeral is always a maximizer of the semantic factor.

Proof. By Assumption 1, $n \in h$ for every $h \in \mathcal{H}_n$, so $\mathbb{1}[n \in h] = 1$ for all h and $S_n(n) = \sum_{h \in \mathcal{H}_n} \theta_h = 1$. For any $m \in D$, $S_n(m) \leq \sum_{h \in \mathcal{H}_n} \theta_h = 1$, since θ sums to one. $\square$

Theorem 1 (Shape). Under Assumptions 1 and 2, S_n is non-increasing as m moves away from n in either direction: for $n \leq m_1 \leq m_2$, $S_n(m_1) \geq S_n(m_2)$, and likewise for $m_2 \leq m_1 \leq n$. Moreover, $S_n(m) = \theta(\{h \in \mathcal{H}_n : m \in h\})$, so S_n changes value only at the boundaries of descriptions and is constant between them.

Proof. Take $n \leq m_1 \leq m_2$ and any $h \in \mathcal{H}_n$ containing m_2 . By centeredness, $n \in h$. By solidity, h contains every value between n and m_2 , m_1 in particular. Every description containing m_2 contains m_1 , hence $\mathbb{1}[m_1 \in h] \geq \mathbb{1}[m_2 \in h]$ for every h . Summing with weights θ_h gives $S_n(m_1) \geq S_n(m_2)$. The case below n is symmetric. For the second claim, $S_n(m)$ is by definition

the total weight of the descriptions containing m ; as m moves, this total changes only when m enters or leaves a description, that is, at a description’s boundary. $\square$

The maximizers of S_n are the values contained in every description, that is, the intersection $\bigcap \mathcal{H}_n$, since only a value satisfying every description collects the entirety of the probability mass θ . If the exact description $\{n\}$ is present in the family, the intersection is $\{n\}$ and the numeral is the unique maximizer: the bare numeral has p_n in its family and so has a spike at n . If it is not, the intersection is a convex set containing n over which S_n forms a flat plateau. Whenever the resolution family is nested by inclusion, as scale-generated families are, this set is the narrowest description in the family: *approximately* n removes p_n , and its factor is flat across the narrowest loose description. These results govern the semantic factor S_n . The posterior is proportional to $\rho(m) \cdot S_n(m)$, and a sufficiently skewed quantity prior ρ can move the posterior’s maximum away from n .

The roundness and magnitude effects rest on two empirical premises about which descriptions a context supplies. By Theorem 1, $S_n(n + d)$ is the total weight of the descriptions of n that extend at least d beyond n ; comparing two numerals’ semantic factors is therefore comparing how far their families reach.

Assumption 3 (Roundness). For a round numeral n' and a non-round numeral n of comparable size, the probability mass assigned to descriptions extending at least a distance d beyond n' is greater than or equal to that of descriptions extending at least d beyond n , for all $d \geq 0$.

Assumption 4 (Magnitude). Given a numeral of larger magnitude n' and a smaller numeral n , the probability mass assigned to descriptions extending at least a distance d beyond n' is greater than or equal to that of descriptions extending at least d beyond n , for all $d \geq 0$.

Under either assumption, Theorem 1 gives $S_{n'}(n' + d) \geq S_n(n + d)$ for every $d \geq 0$, and likewise below: at every distance from the numeral, the semantic factor of the round/larger numeral is at least as high as that of the non-round/smaller one.

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